

EE 660 Power Management IC Design

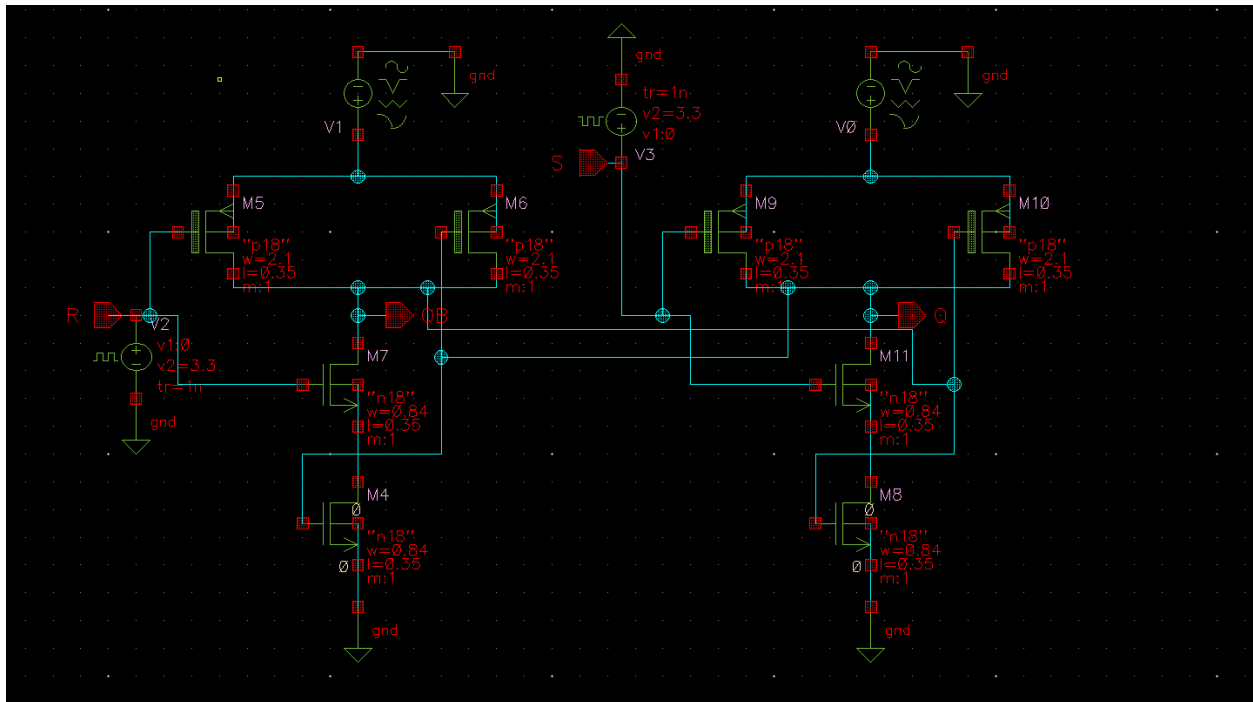
Design of a closed-loop voltage-mode buck converter

SAHASRITH BOOTLA (23110064)

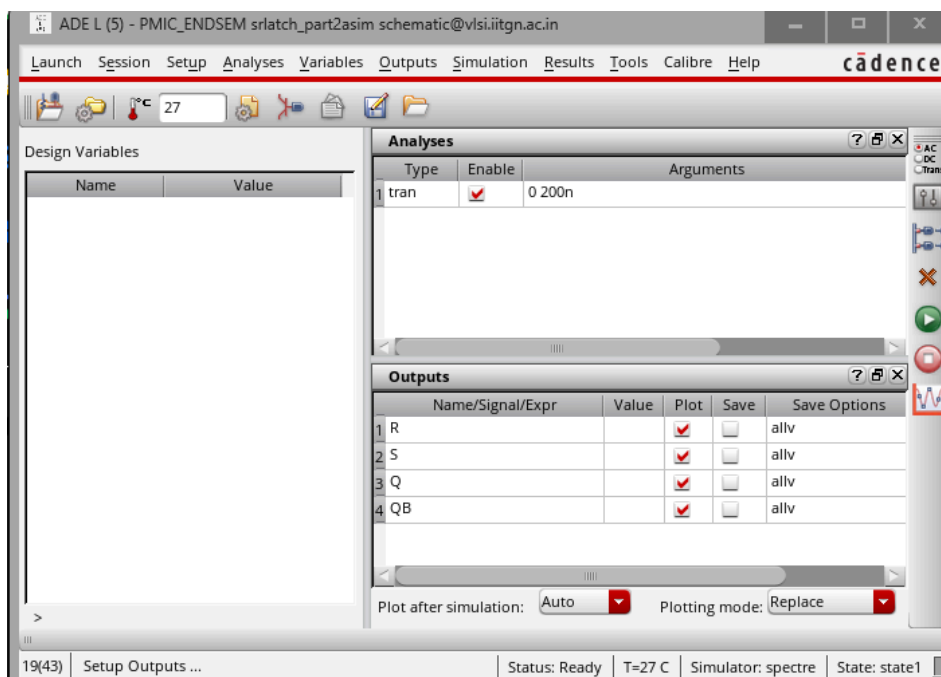
Part 2 A : Comparator design + SR Latch

Design and test comparator separately with $t_{\text{delay}} < 10\text{ns}$ (1% of T_{sw})

1. SR LATCH



ADL

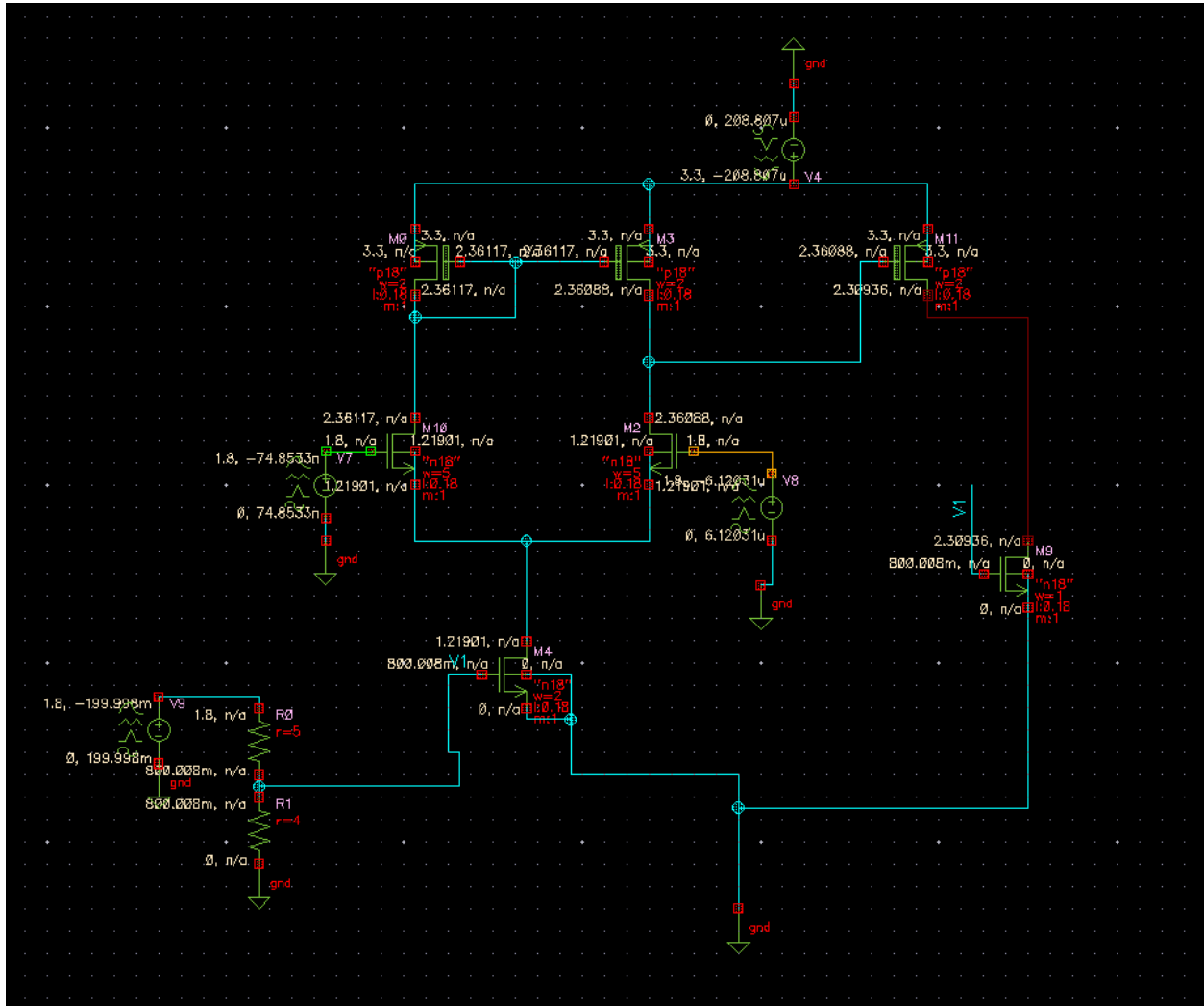


OUTPUT

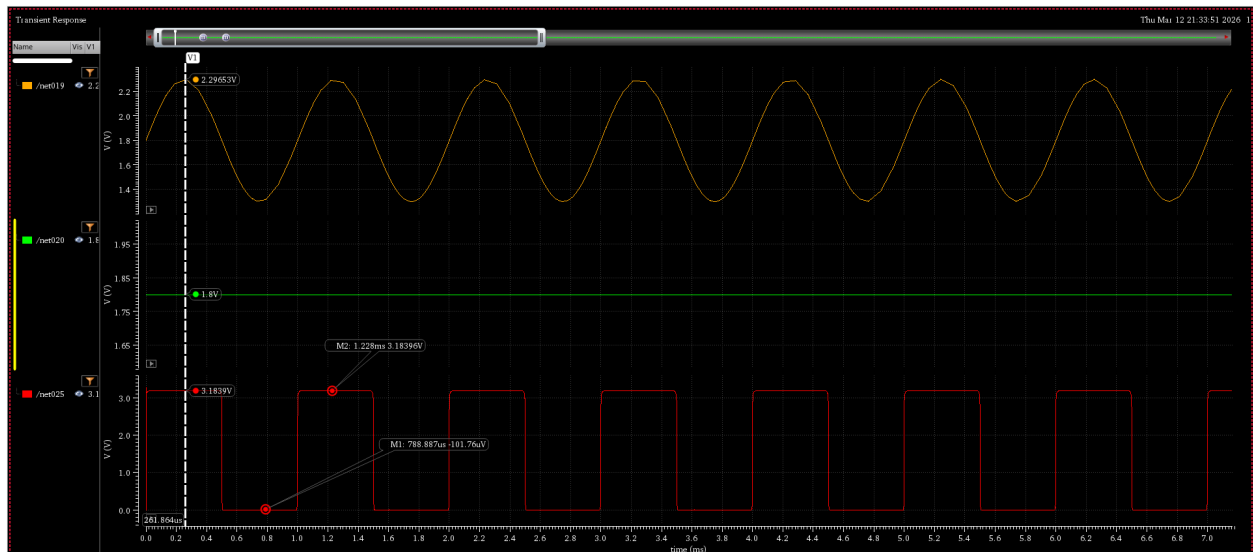


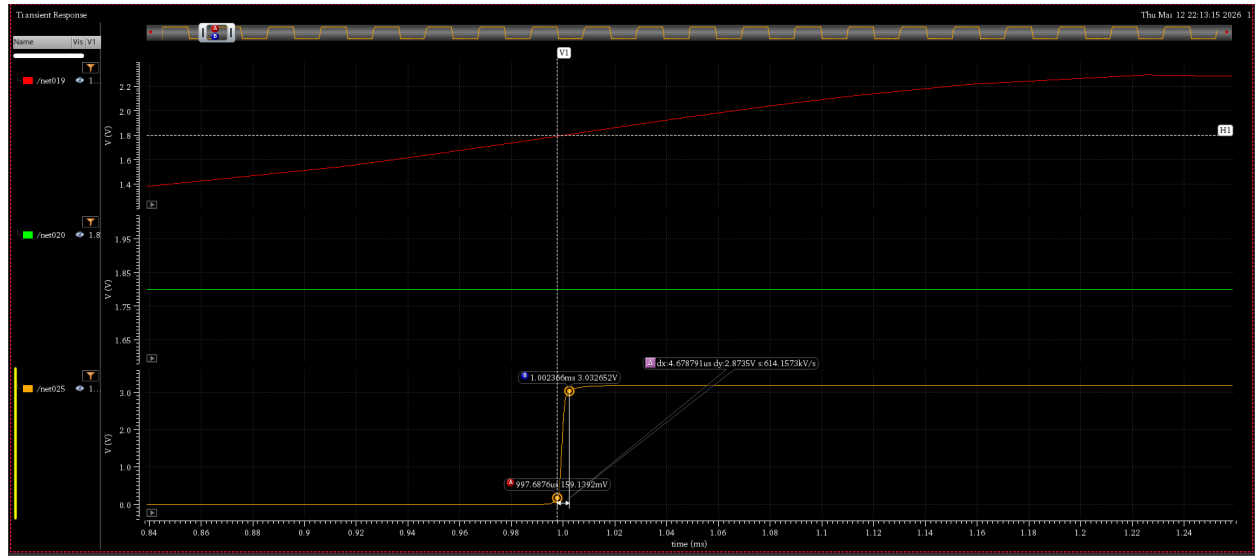
Transistor	Type	W (μm)	L (μm)
M5	PMOS	2.1	0.35
M6	PMOS	2.1	0.35
M9	PMOS	2.1	0.35
M10	PMOS	2.1	0.35
M7	NMOS	0.84	0.35
M11	NMOS	0.84	0.35
M4	NMOS	0.84	0.35
M8	NMOS	0.84	0.35

2. COMPARATOR



Note : Transient Voltages and currents at $t = 261.864\mu s$



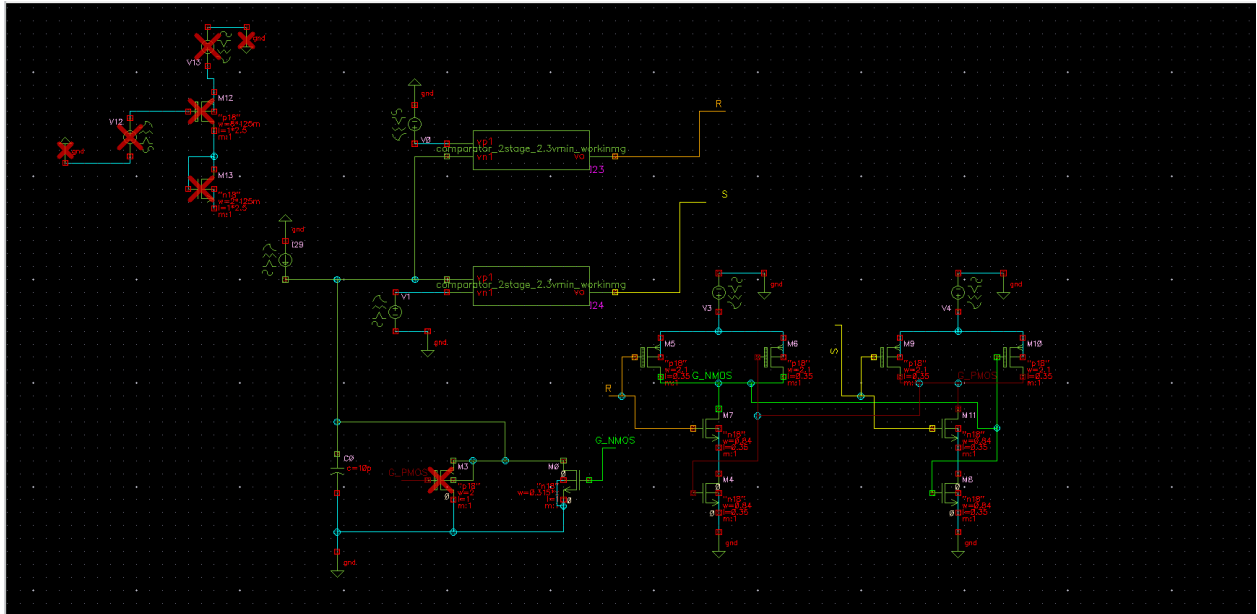


MOSFET	Type	W (μm)	L (μm)
M8	PMOS	2	0.18
M3	PMOS	2	0.18
M11	PMOS	2	0.18
M10	NMOS	5	0.18
M2	NMOS	5	0.18
M4	NMOS	5	0.18
M9	NMOS	1	0.18

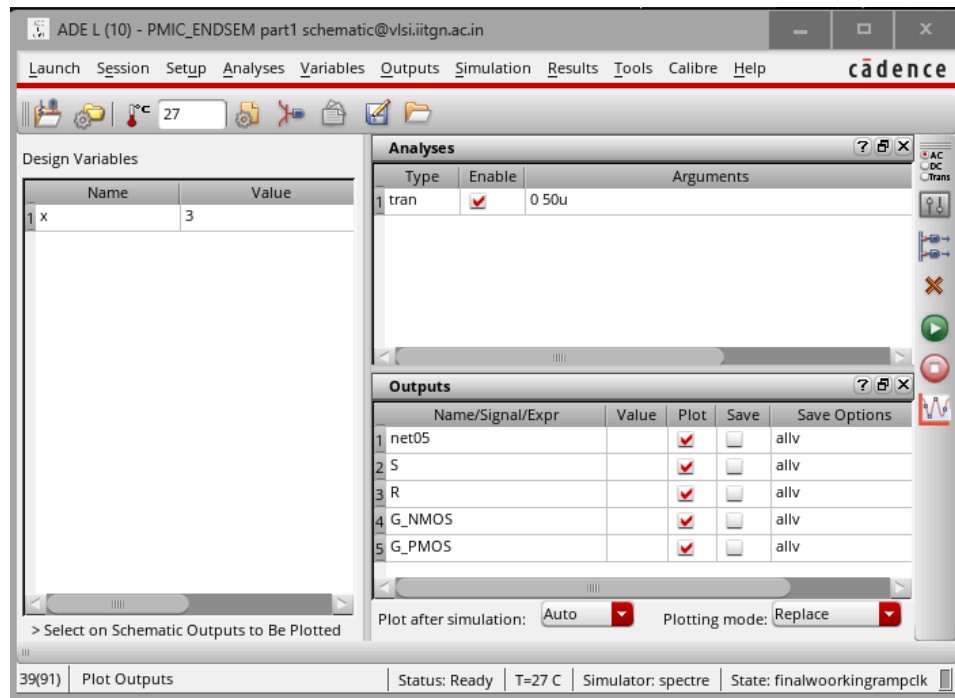
Part 2 B : RMP + CLK circuit

Design the ramp/clock generator (Recall that you will require comparators within the ramp generator circuit as well).

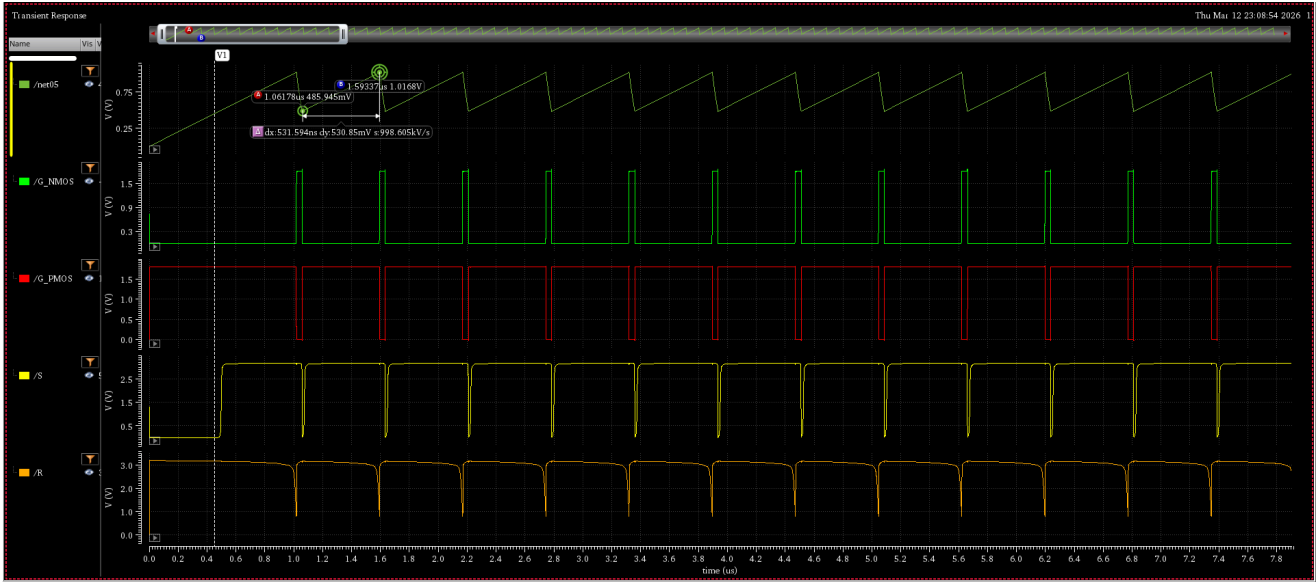
SCHEMATIC



ADL

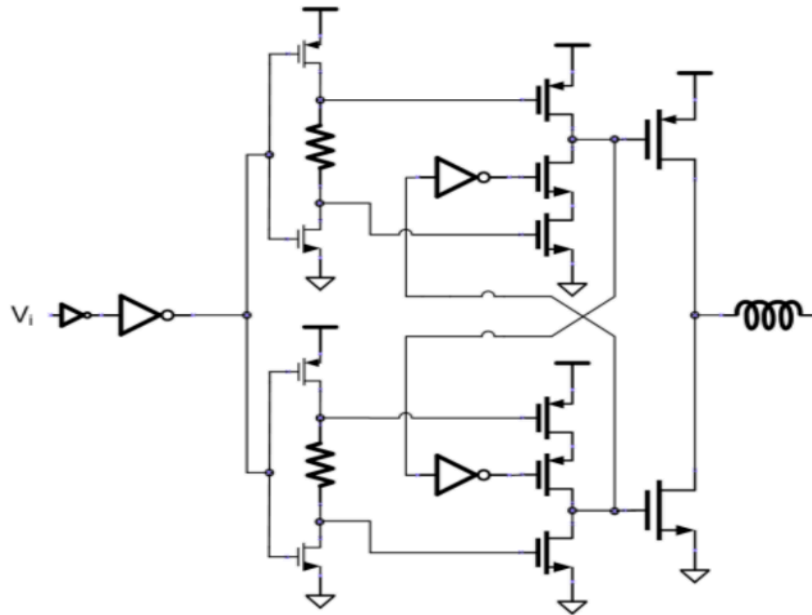


OUTPUT



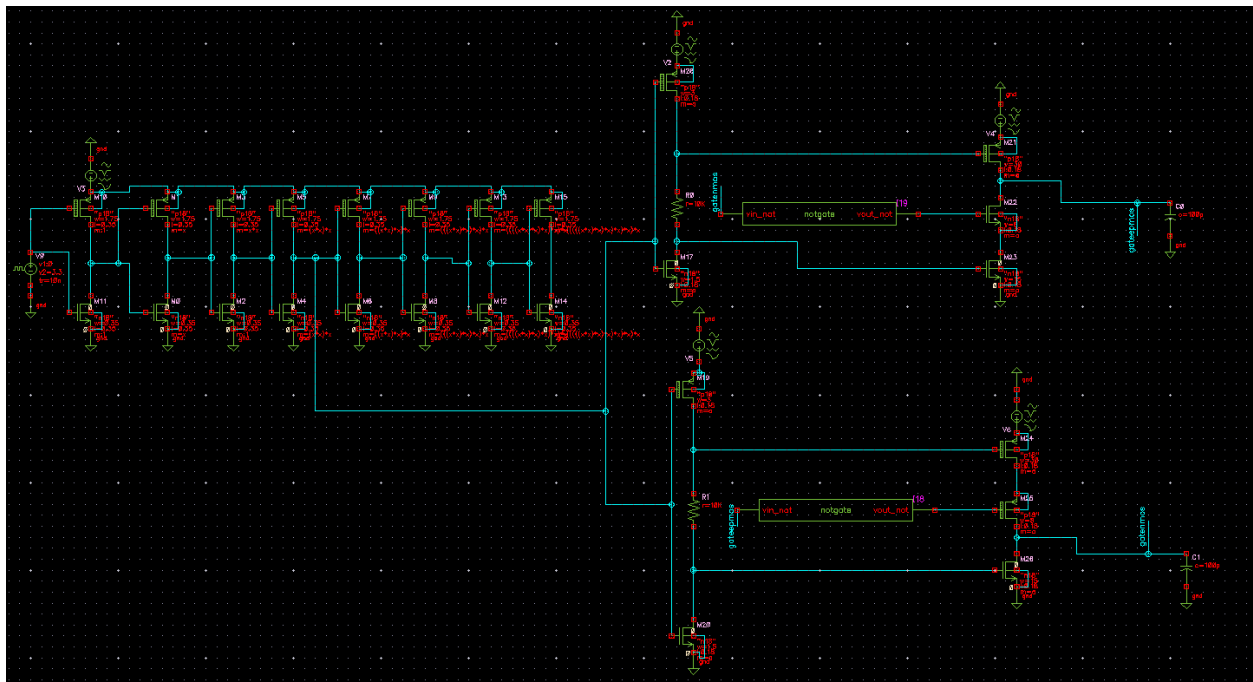
Transistor	Type	W (μm)	L (μm)
M0	NMOS	0.945	1

Part 2 D: Buffer + Break before make driver



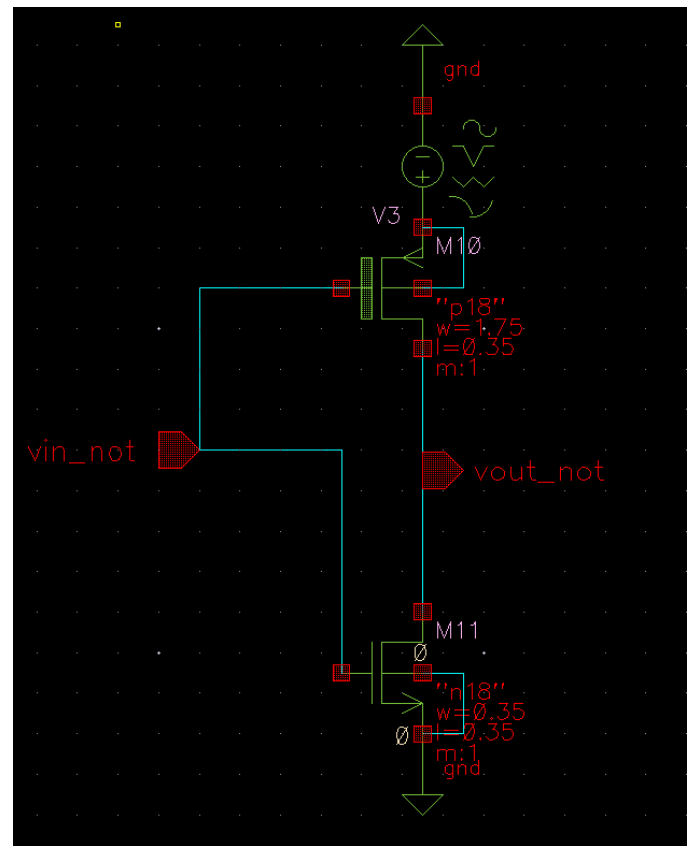
1. Design the Buffer + Break before make driver separately and integrate the PWM modulator
 2. Consider two capacitors each of 100pF for emulating the switching stage.
- Show the plots showing the working with the suitable testbench.

SCHEMATIC

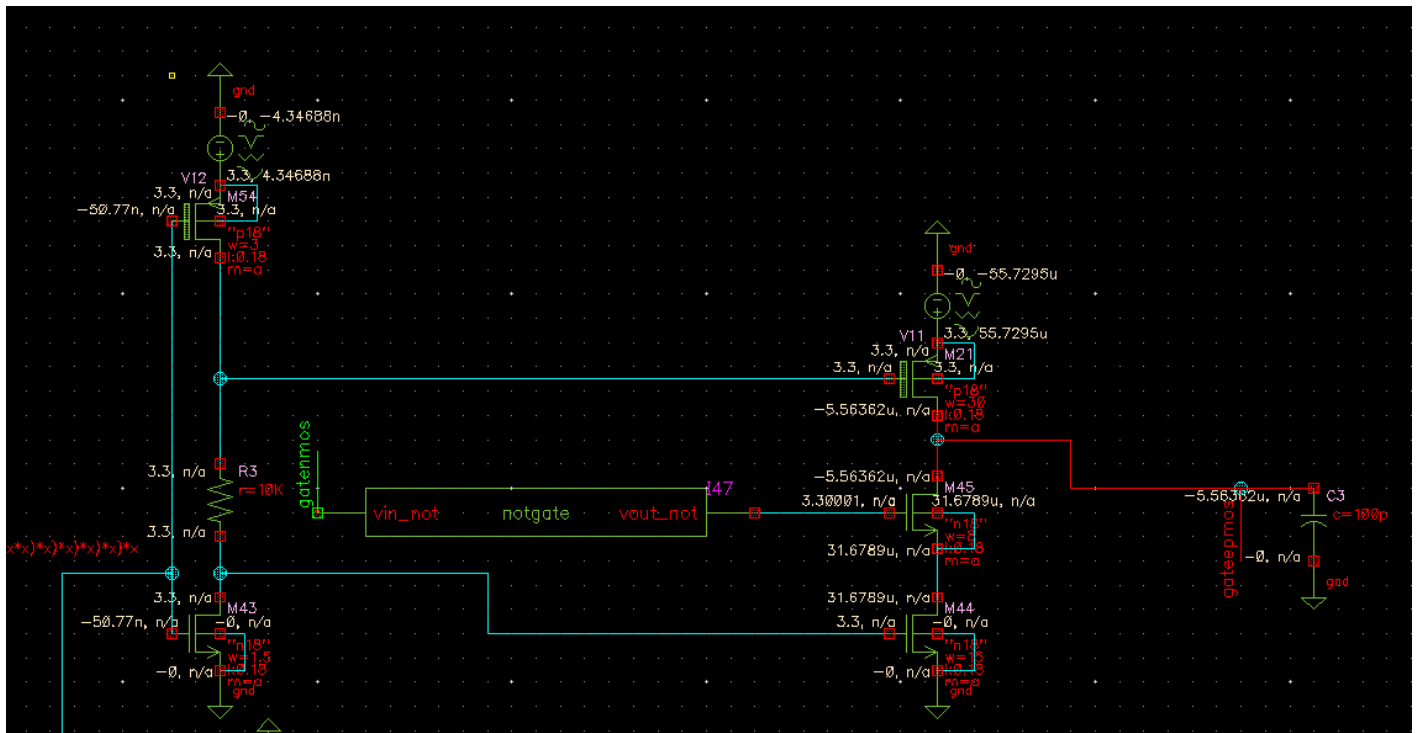


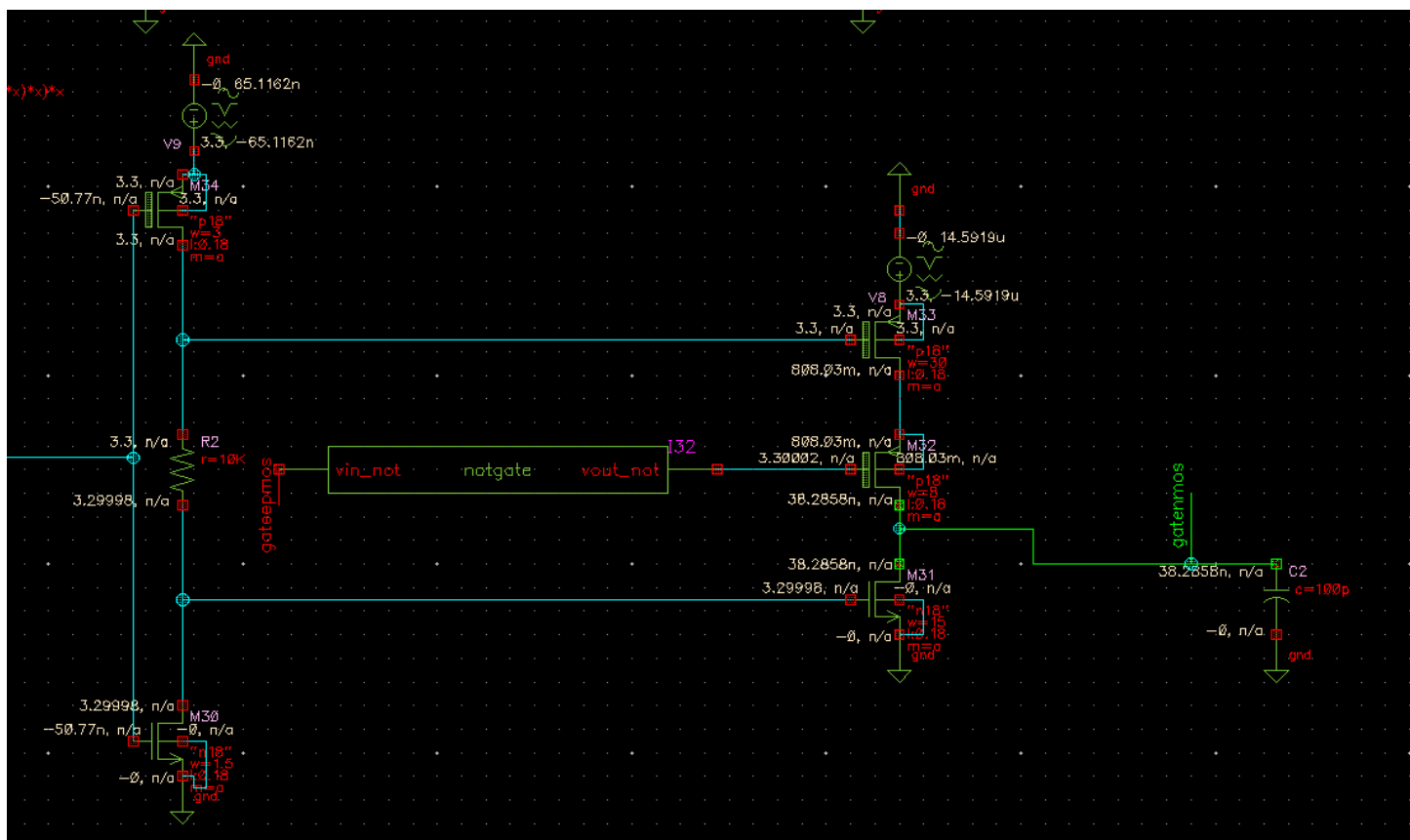
NOT GATE :

MOSFET	Type	W (μm)	L (μm)
M10	PMOS	1.75	0.35
M11	NMOS	0.35	0.35

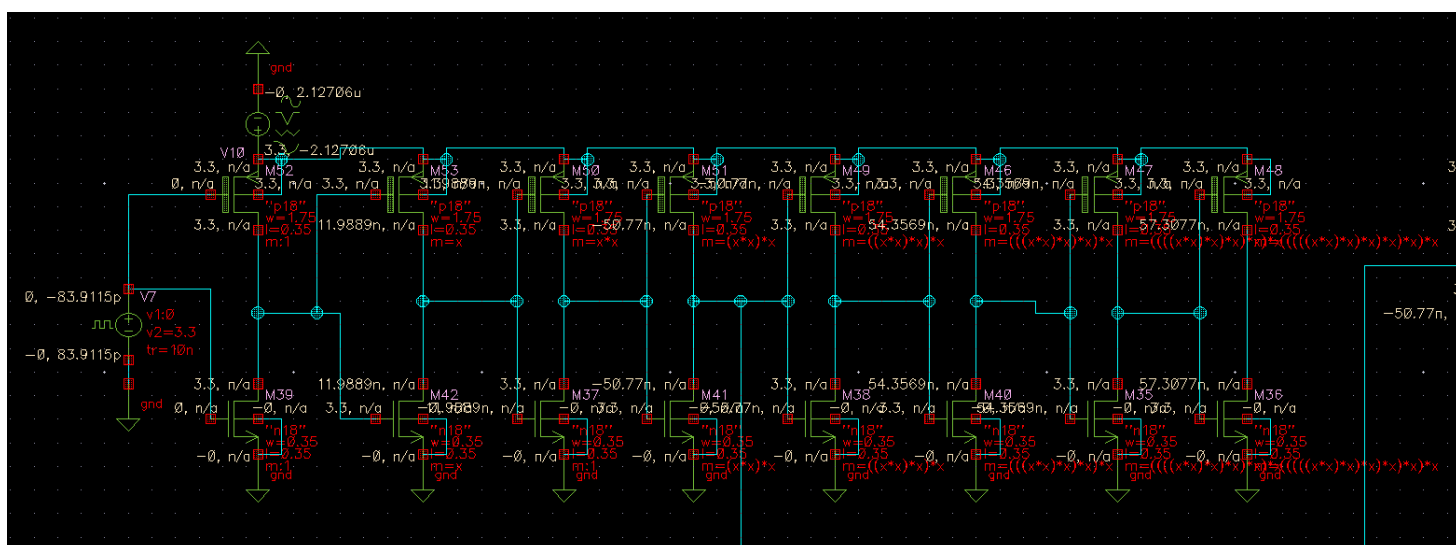


TRANSIENT VOLTAGES,CURRENTS at t = 1.23989us

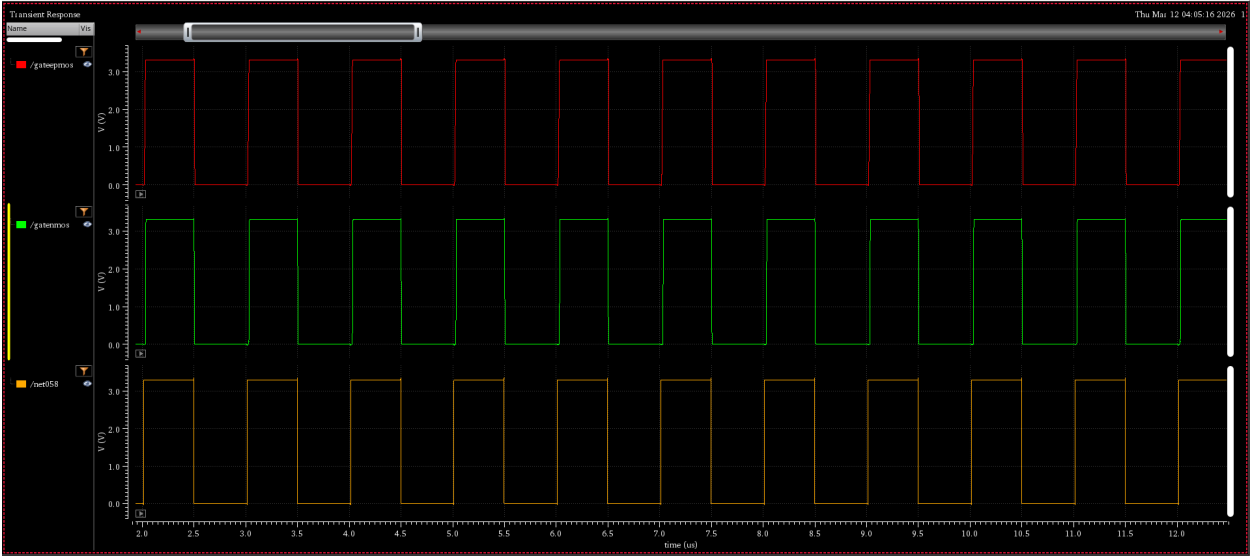




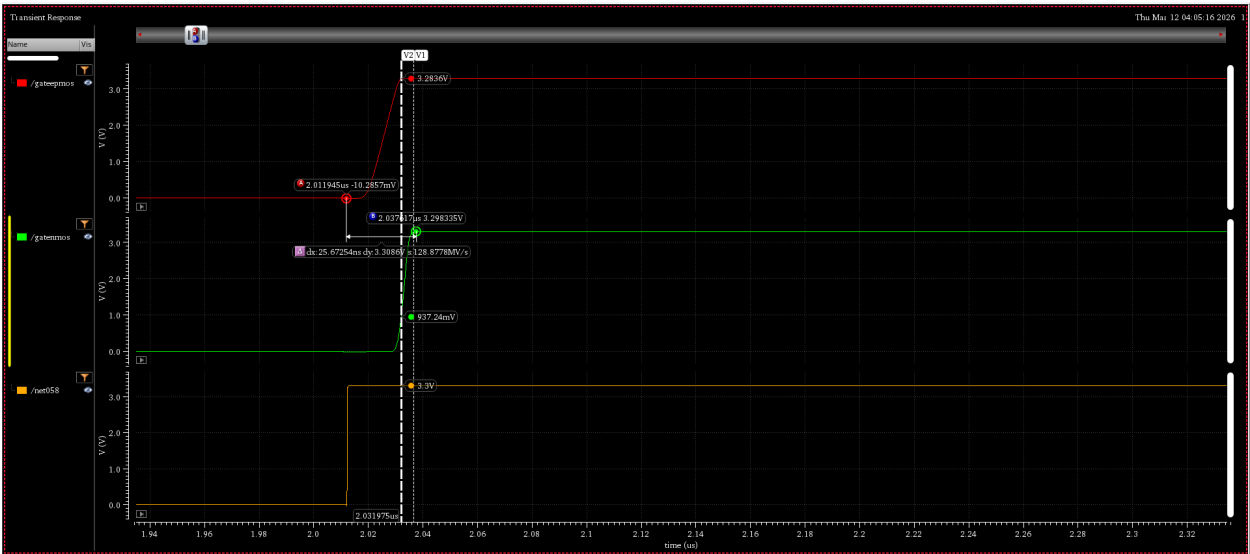
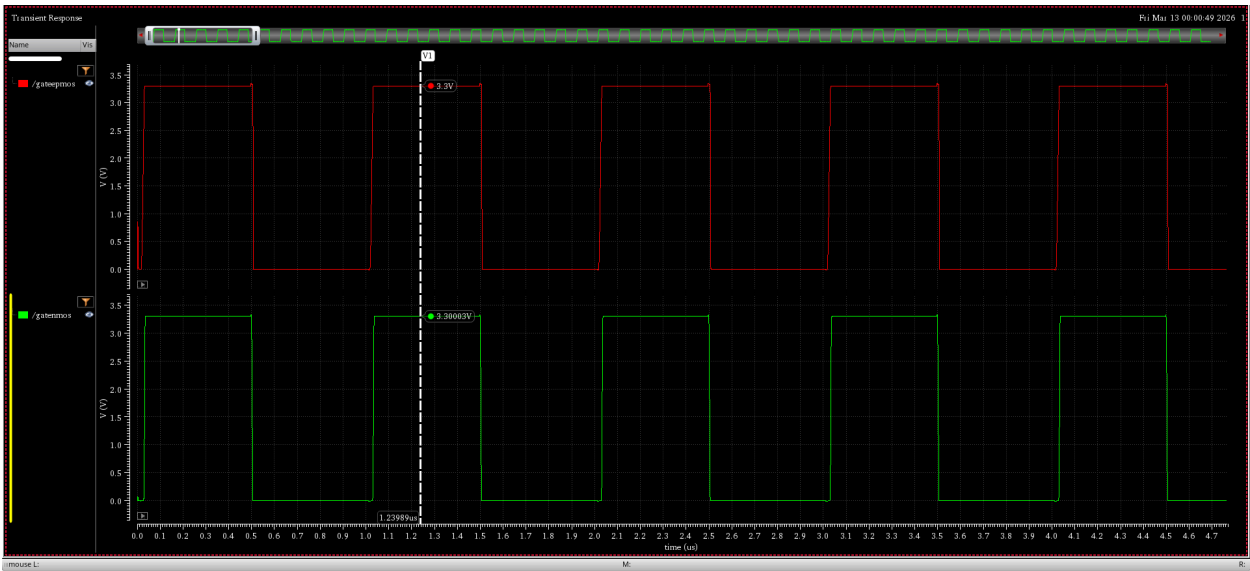
BUFFER



OUTPUT



$t = 1.23989\mu s$



BUFFER SIZING:

Stage 1

MOSFET	Type	W (μm)	L (μm)	Multiplier m
M52	PMOS	1.75	0.35	1
M39	NMOS	0.7	0.35	1

Stage 2

MOSFET	Type	W (μm)	L (μm)	Multiplier m
M53	PMOS	1.75	0.35	5
M42	NMOS	0.7	0.35	5

Stage 3

MOSFET	Type	W (μm)	L (μm)	Multiplier m
M50	PMOS	1.75	0.35	25
M37	NMOS	0.7	0.35	25

Stage 4

MOSFET	Type	W (μm)	L (μm)	Multiplier m
M51	PMOS	1.75	0.35	125
M41	NMOS	0.7	0.35	125

Break-Before-Make Driver stage sizing

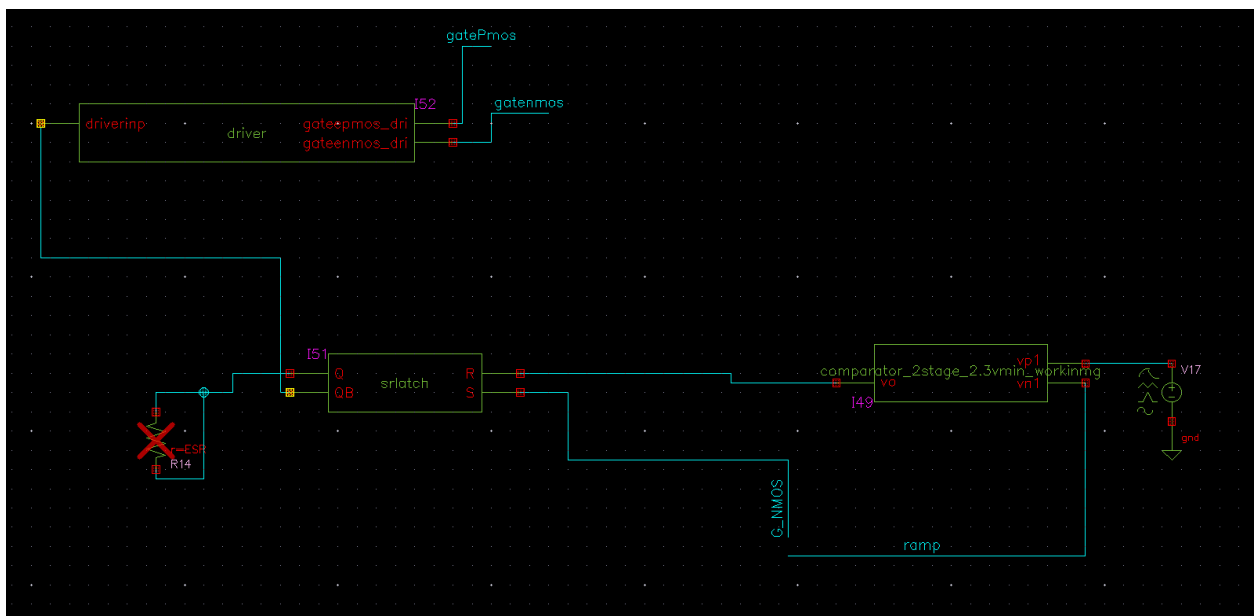
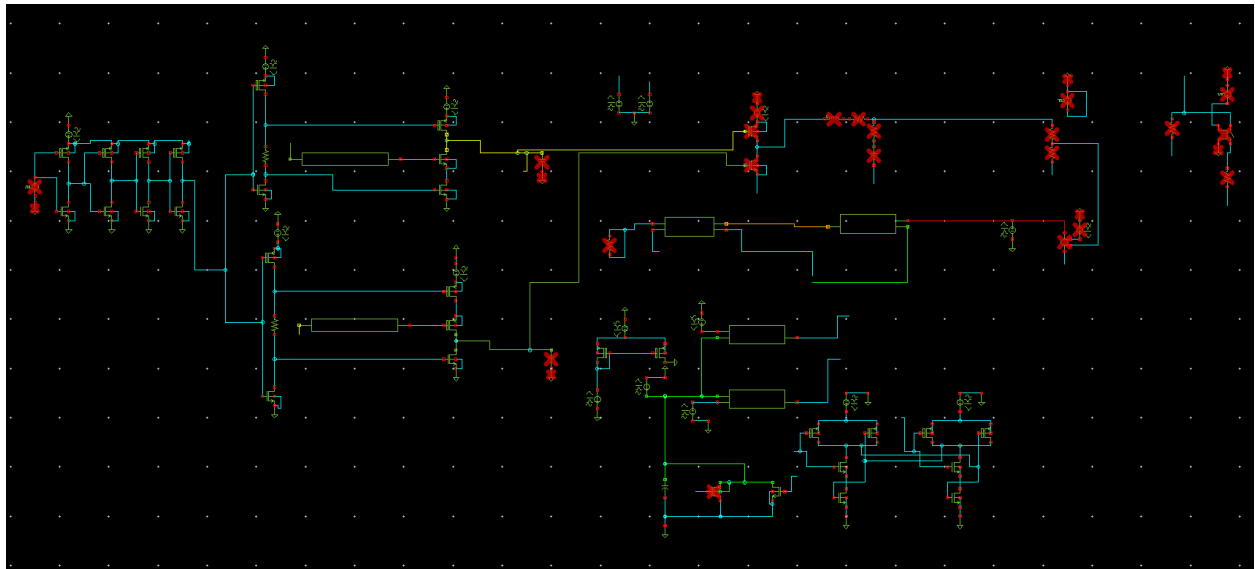
MOSFET	Type	W (μm)	L (μm)	Multiplier (m)
M54, M34	PMOS	3	0.18	100
M43, M30	NMOS	1.5	0.18	100
M21, M33	PMOS	30	0.18	100
M45, M32	PMOS	8	0.18	100
M44, M31	NMOS	155	0.18	100

Part 2 C: Integration of RMP + CLK circuit, Comparator design + SR Latch

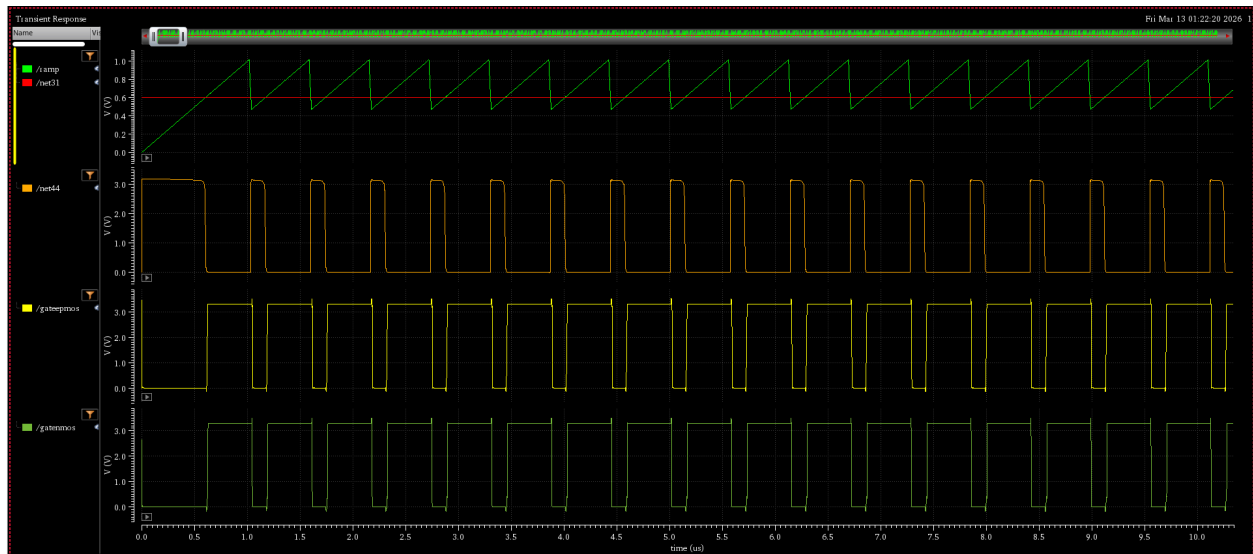
Integrate them to complete the PWM modulator.

Apply a single DC voltage (emulating output of error amplifier) to the comparator which is varied between 0.1 VH and 0.9 V and show change in duty cycle.

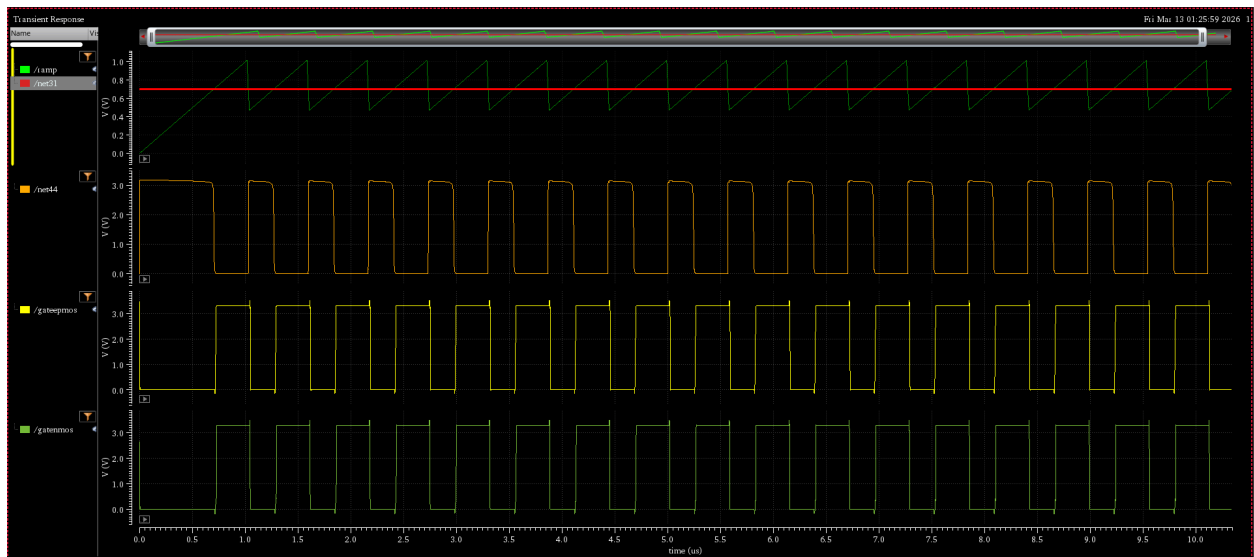
SCHEMATIC



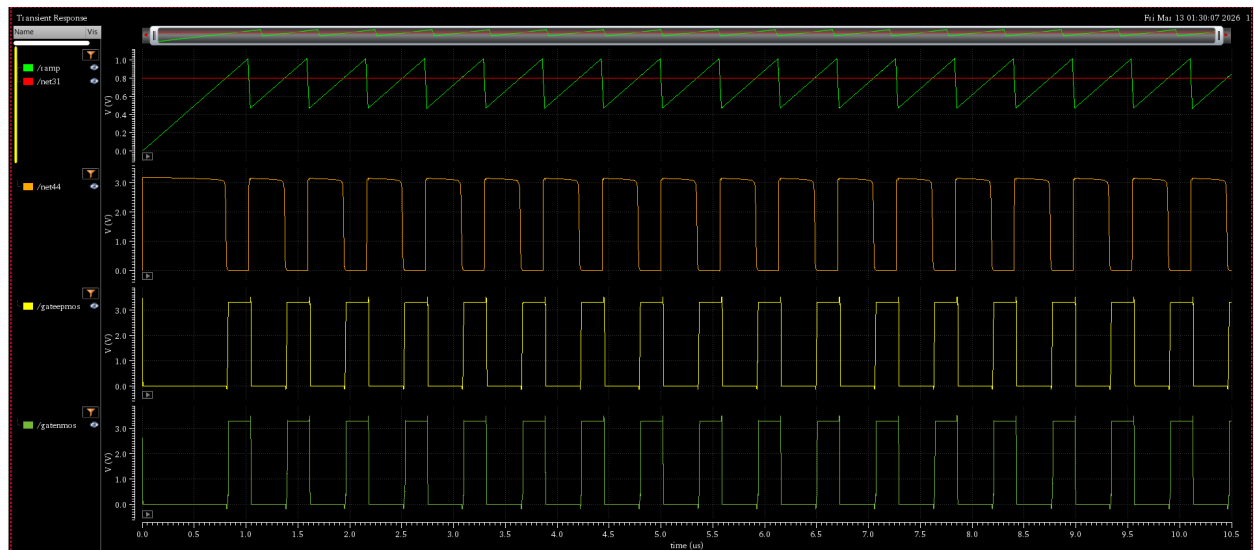
$V_{comp} = 0.6$



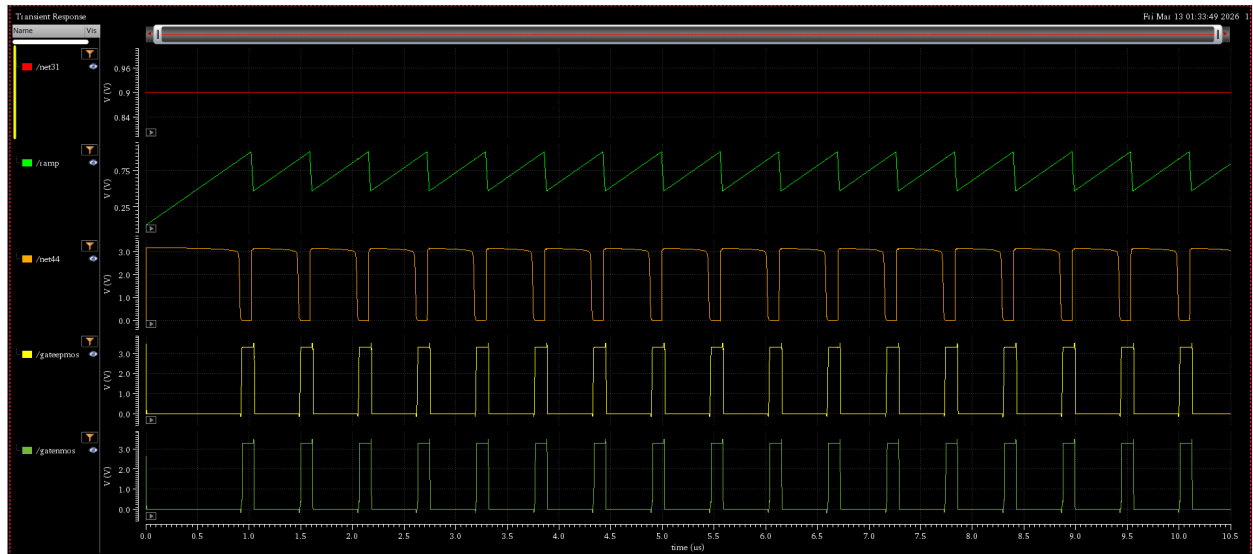
$V_{comp} = 0.7$



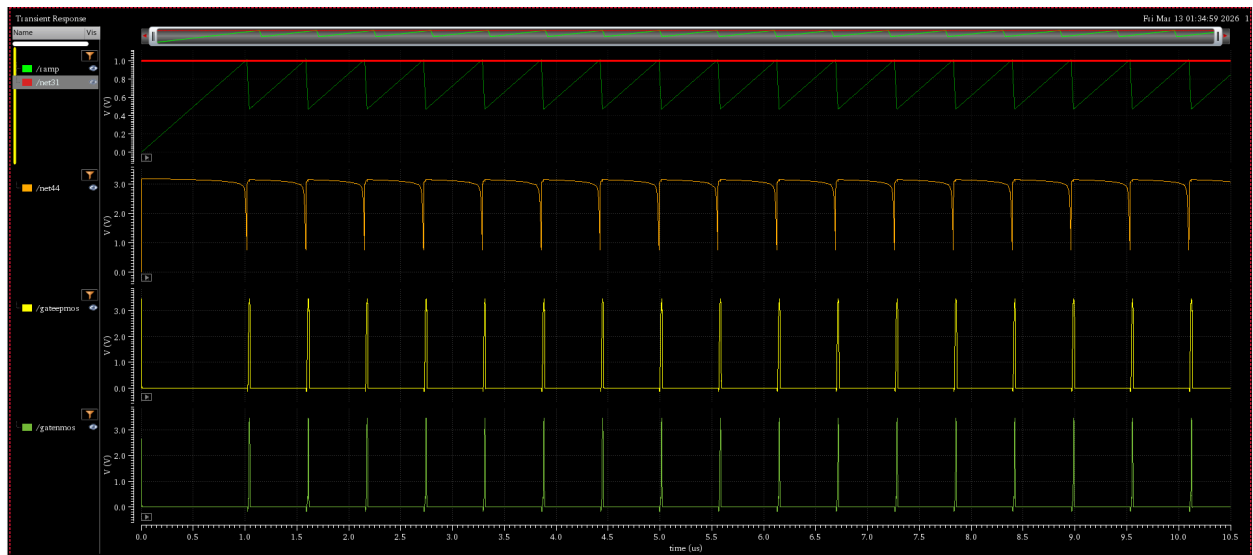
$V_{comp} = 0.8$



$V_{comp} = 0.9$



$V_{comp} = 1$



ADL

ADE L (13) - PMIC_ENDSEM part2C_comparator+latch schematic@vlsi.iitgn.ac.in

Launch Session Setup Analyses Variables Outputs Simulation Results Tools Calibre Help

Design Variables

Name	Value
1 mul_mos	1K
2 a	100
3 gain_error	1K
4 x	5
5 rload	18
6 vdd	3.3
7 Rloss	50m
8 ESR	20m
9 L	6.8u
10 Cout	4.41u
11 T	1u

Analyses

Type	Enable	Arguments
1 tran	☒	0 10.5u moderate

Outputs

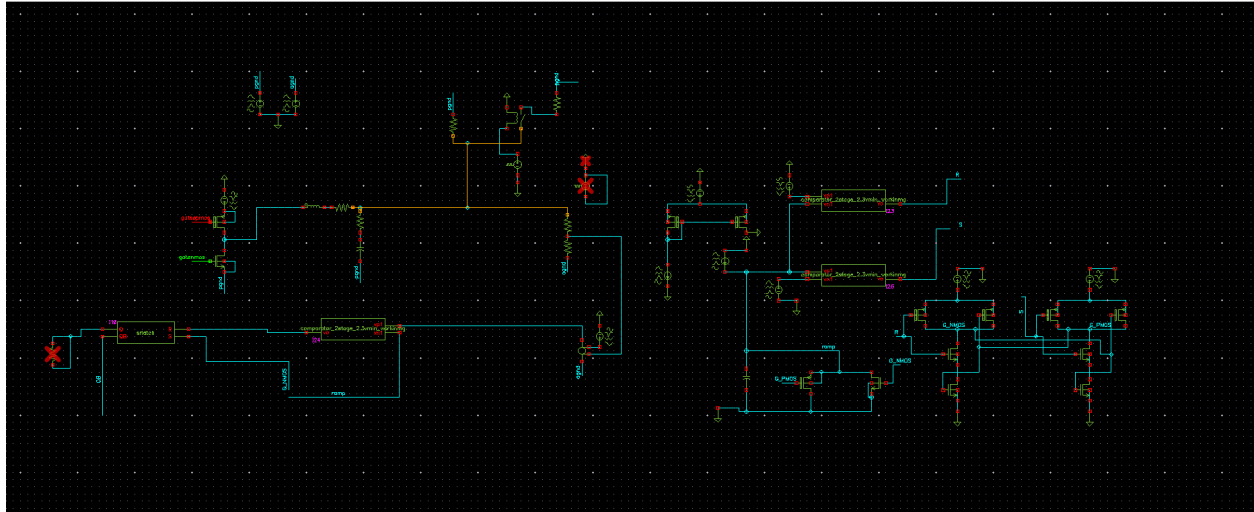
Name/Signal/Expr	Value	Plot	Save	Save Options
1 net31		☒	☒	yes
2 ramp		☒	☒	yes
3 net44		☒	☒	yes
4 gatepmos		☒	☒	yes
5 gatenmos		☒	☒	yes

Plot after simulation: Auto Plotting mode: Replace

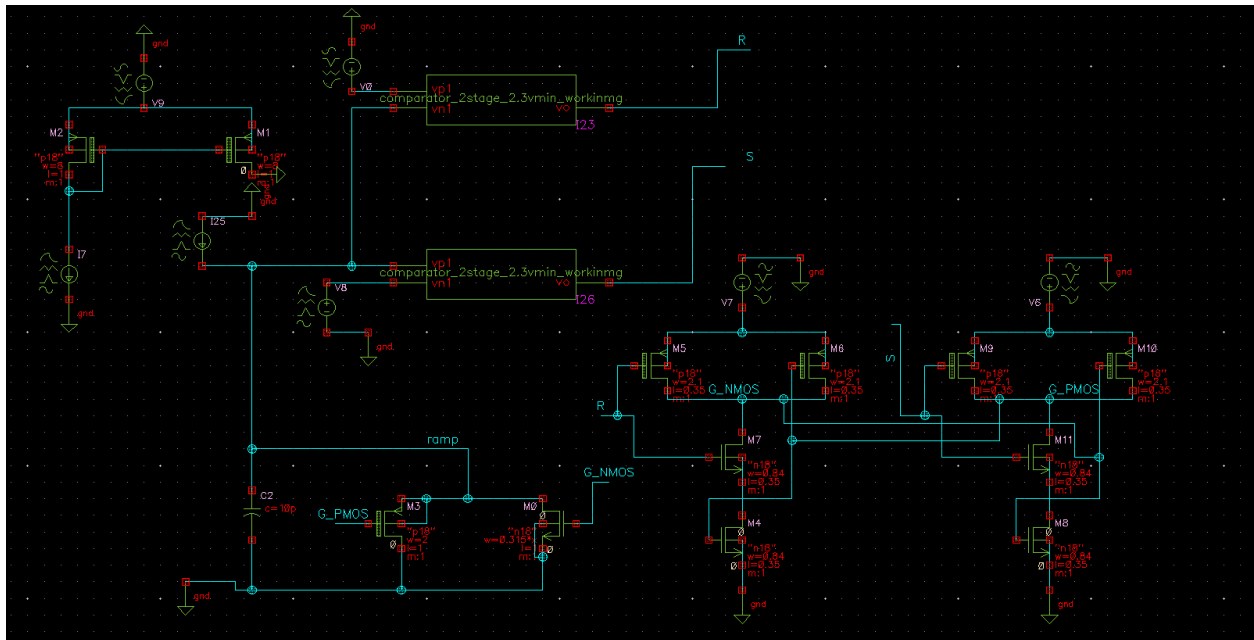
49(121) Netlist and Run Status: Ready T=27 C Simulator: spectre State: state1

Part 2 E: Replace ideal Opamp and compensation in Part 1 B. (PWM generator and power stage are ideal)

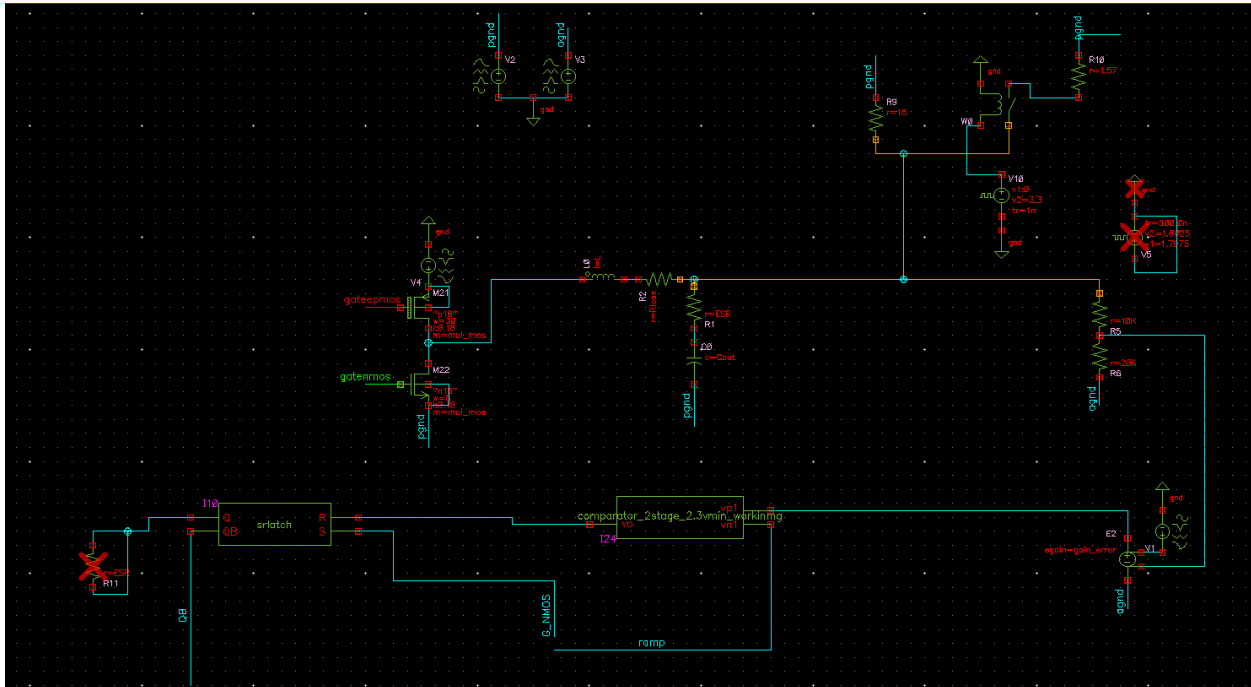
SCHEMATIC {first checking with ideal error ampl}



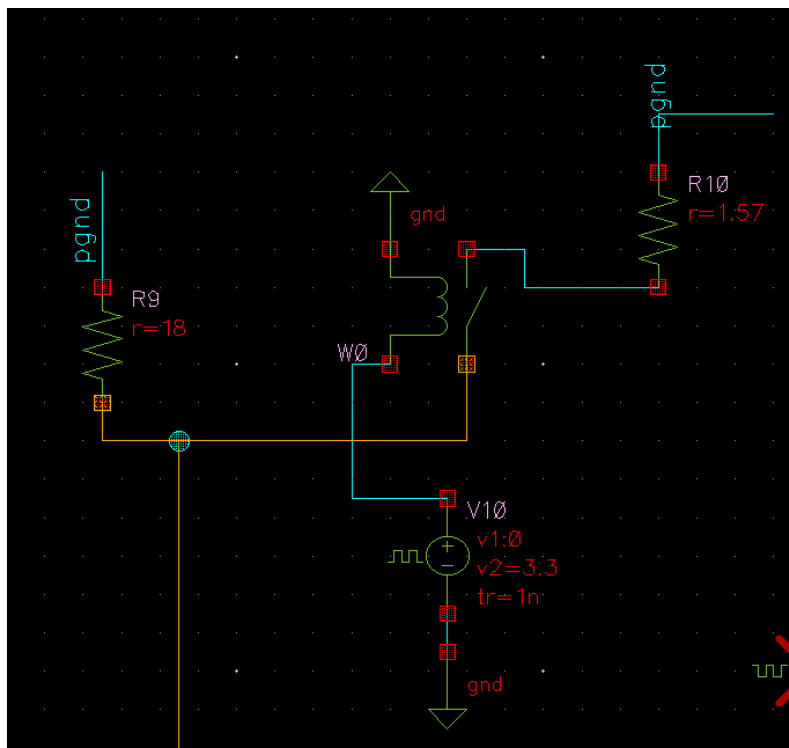
RAMP+CLK



POWER STAGE



LOAD BLOCK



OUTPUT

